

4.0 Milestone Assessment: Network Incident Investigation Report

Introduction

As the lead Network Administrator, I was tasked with investigating multiple network issues affecting users across the organization. Employees reported:

1. Slow access to network resources and the internet,
2. Intermittent network disconnections, and
3. Unusual traffic patterns suggesting a potential security breach.

Because these symptoms could be related, it was essential to follow a structured troubleshooting methodology rather than making assumptions or immediately changing network configurations. Effective network troubleshooting begins by understanding the problem before attempting a solution. The steps in this process include:

1. Gather information,
2. Determine incident scope,
3. Select appropriate diagnostic tools and protocols,
4. Collect and analyze evidence,
5. Implement corrective actions, and
6. Verify that the problem has been resolved.

Following a systematic approach minimizes downtime, reduces unnecessary changes, and increases the likelihood of identifying the true root cause instead of simply treating the symptoms.

The objectives of this investigation were to identify the root causes of the reported issues, restore normal network performance and connectivity, secure the network against potential threats, and verify that critical network services continue to operate correctly.

Team members investigated each aspect of the incident using appropriate diagnostic tools and protocols, then combined their findings to develop a complete understanding of the situation. This collaborative, evidence-based approach enabled the team to implement effective solutions, validate the results, and establish recommendations for continuous monitoring and future prevention.

Investigation Summary

Issue	Diagnostic Tools & Protocols	Key Findings	Resolution
Slow Network Performance	Wireshark, SNMP	Network congestion and high interface utilization	Optimized network traffic flow and reduced congestion
Intermittent Connectivity	Ping, Traceroute, IPv4 configuration	Incorrect network configuration causing communication failures	Connectivity restored after correcting configuration issues
Potential Security Breach	Firewall Logs, IDS, IPS, Nmap	Suspicious traffic and unauthorized connection attempts detected	Isolated affected systems and strengthened monitoring

Investigation

Slow Network Performance

Employees reported slow access to network resources and the Internet. The objective of this investigation was to determine whether the slowdown was caused by excessive traffic, overloaded network devices, or another underlying network issue.

The **Network Analyst** investigated reports of slow network performance using Wireshark and SNMP (simple network management protocol). Wireshark captured network traffic to identify packet loss, excessive broadcasts, retransmissions, and abnormal traffic patterns.

SNMP was used to monitor the health and utilization of network devices, including CPU usage, memory consumption, interface statistics, and bandwidth utilization. Analysis revealed network congestion caused by excessive traffic and high interface utilization. After identifying the bottleneck, traffic was optimized by reducing unnecessary network traffic and balancing utilization across network resources. Follow-up monitoring confirmed that normal network performance had been restored.

Intermittent Connectivity

Employees reported intermittent network disconnections. The investigation examined whether connectivity issues were related to hardware failures, configuration errors, DHCP issues, routing problems, or unstable network links.

The **Systems Engineer** investigated intermittent connectivity using ping and traceroute. Ping verified communication between devices while measuring latency and packet loss. Traceroute identified the location where communication failures occurred by displaying each router along the transmission path.

The engineer also verified client network configuration, including IPv4 addressing, subnet mask, default gateway, DNS settings, and DHCP lease information, to ensure proper network addressing. Correcting the identified addressing and configuration issues restored reliable connectivity and eliminated intermittent network interruptions.

Potential Security Breach

Unusual traffic patterns were detected, requiring investigation into whether the activity was caused by normal network behavior, misconfiguration, unauthorized access, malware activity, or another potential security concern.

The **Security Specialist** investigated unusual network traffic by reviewing firewall logs, analyzing alerts generated by the organization's Intrusion Detection System (IDS) and Intrusion Prevention System (IPS), as well as performing network discovery and port scanning with Nmap. Firewall logs identified suspicious connection attempts, IDS detected suspicious activity, IPS automatically blocked unauthorized traffic, and Nmap identified active hosts, open ports, and exposed network services.

The affected systems were isolated, malicious traffic was blocked, security policies were updated, and continuous monitoring was implemented. User accounts and affected devices were also reviewed to ensure no unauthorized access remained after the incident. Critical network services, including DHCP, DNS, and IPv6 services, were verified to ensure they continued operating correctly after the security response.

Network Design Considerations

Although the scenario doesn't specify the organization's network topology or architecture, network administrators must understand the strengths and limitations of common designs before recommending improvements or implementing changes.

Network Topologies

Topology	Advantages	Disadvantages
Star	Easy to troubleshoot, scalable, and isolates individual device failures	Central switch is a single point of failure
Bus	Simple and inexpensive	Backbone failure disrupts the entire network and limits scalability
Ring	Predictable communication with minimal collisions	A single break can interrupt communication without redundancy
Mesh	Excellent redundancy and fault tolerance	Expensive and complex to deploy and maintain
Hybrid	Flexible and scalable by combining multiple topologies	More complex to design and troubleshoot

Network Architectures

Architecture	Advantages	Disadvantages
Client-Server	Centralized authentication, security, management, and resource sharing	Server failures can affect many users without redundancy
Peer-to-Peer	Simple and inexpensive for small networks	Difficult to manage, secure, and scale

Network Types

Network Type	Description
LAN	Connects devices within a limited geographic area such as a home, school, or business.
WLAN	A wireless LAN providing network access using Wi-Fi.
WAN	Connects multiple LANs across large geographic distances.
MAN	Connects networks across a city or metropolitan area.
PAN	Connects personal devices over short distances using technologies such as Bluetooth.

Understanding these network designs enables administrators to isolate failures more efficiently, improve network performance, and design resilient networks that support business operations.

Lessons Learned and Future Recommendations

This investigation reinforced that effective network troubleshooting requires a structured, evidence-based approach. One of the most important lessons learned was that unusual network traffic should never be ignored, as it may indicate malware, unauthorized access, or other malicious activity. By using multiple security tools, including firewall logs, Intrusion Detection Systems (IDS), Intrusion Prevention Systems (IPS), Wireshark, and Nmap, the team identified suspicious activity, analyzed network traffic, discovered exposed services, and successfully contained the threat.

Another important lesson was that security incidents can produce symptoms similar to network congestion or hardware failures. Collecting and analyzing evidence before making configuration changes allowed the team to identify the true root cause while avoiding unnecessary modifications to the network.

To improve future incident response, the organization should implement a Security Information and Event Management (SIEM) platform to centralize log collection from firewalls, IDS/IPS systems, servers, and endpoints. Correlating security events across multiple systems would improve network visibility, enable earlier threat detection, and help identify coordinated attacks before they significantly impact network operations.

The team should also:

- Continue monitoring network performance using SNMP and Wireshark.
- Regularly review firewall logs and IDS/IPS alerts, and perform periodic Nmap scans to identify unauthorized devices, open ports, and exposed services.
- Verify IPv4 addressing, DHCP leases, DNS settings, and default gateway configurations whenever connectivity issues occur.
- Maintain current network documentation, establish performance baselines, and keep network devices updated with security patches.
- Develop and regularly practice an incident response plan so team members can respond quickly, consistently, and effectively during future network and security incidents.

Implementing these recommendations will strengthen the organization's ability to prevent, detect, investigate, and respond to future network performance and security incidents while maintaining a secure, reliable, and resilient network environment.

Conclusion

This investigation demonstrated the importance of following a structured troubleshooting methodology when responding to network incidents. By selecting the appropriate tools and protocols, gathering evidence, verifying network configurations, and collaborating as a team before implementing changes, the root causes of the performance, connectivity, and security issues were accurately identified and resolved.

The experience reinforced that successful network administration depends on logical problem-solving, effective communication, accurate analysis, and continuous monitoring. Applying these best practices not only restores network operations but also strengthens the organization's ability to prevent, detect, and respond to future incidents. This investigation demonstrated that a systematic, proactive approach is essential for maintaining a secure, reliable, and high-performing network. Ultimately, successful network administration is not simply about repairing problems, it's about understanding how networks operate, making informed decisions based on evidence, and continuously improving the network to prevent future incidents.

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